YoloV8 Based Novel Approach for Object Detection on LiDAR Point Cloud

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***Abstract*—Object detection serves as an important perception task for both autonomous vehicles and Advanced Driver Assistance Systems (ADAS). While object detection in camera images has been extensively studied, tackling this task with Light Detection and Ranging (LiDAR) data presents unique challenges due to its inherent sparsity. This study introduces a pioneering approach for LiDAR-based object detection, wherein LiDAR point cloud data is ingeniously transformed into a pseudo-RGB image format, subsequently fed into the YOLOv8 network originally designed for camera-based object detection. Trained and rigorously evaluated on the KITTI dataset, our method demonstrates outstanding performance, achieving an impressive mean Average Precision (mAP) of more than 86%. The model was also tested on some point clouds from TiHAN IITH Autonomous Navigation Dataset (TIAND). This remarkable result highlights the efficacy of the proposed approach in harnessing LiDAR data for robust object detection, thus contributing to the advancement of perception capabilities in autonomous driving and ADAS applications.**

***Index Terms*—YOLOv8, Point cloud, BEV, LiDAR, Object detection, ADAS**

I. Introduction

LiDAR-based object detection is pivotal in advancing autonomous vehicles and ADAS due to its unique capabilities and advantages. Unlike traditional camera-based systems, LiDAR sensors offer precise depth information, allowing for accurate 3D mapping of the surrounding environment. This depth data enables robust object detection even in challenging conditions where visual cues may be limited, such as low light [1], glare, or adverse weather. By providing accurate spatial localization of objects, LiDAR enhances critical functions like collision avoidance and path planning, contributing to safer and more efficient navigation in complex traffic scenarios.

Furthermore, LiDAR's ability to generate dense point clouds facilitates precise object segmentation and classification, improving the accuracy of identifying various objects, including vehicles, pedestrians, cyclists, and obstacles. This capability is particularly crucial for ensuring the safety of vulnerable road users and enhancing overall situational awareness of autonomous systems. Additionally, the integration of LiDAR data with other sensor modalities, such as cameras and radar, enhances redundancy and reliability in perception systems. By leveraging multiple sources of information, autonomous vehicles can make more informed decisions and mitigate the risk of single-point failures, ultimately enhancing overall system robustness and safety. Due to its precise depth sensing capabilities and robust performance in various environmental conditions, LiDAR is widely utilized by researchers in the field of ADAS ([2], [3], [4]).

This paper proposes a novel approach for object detection on LiDAR data. In this approach, Bird's Eye View (BEV) images are created from each point cloud frame. Object detection on these images is performed using the YoloV8 model. The proposed object detection model not only improves the accuracy but is also suitable for real-time applications.

This paper's subsequent portions are organized as follows. Section II contains the literature review of related research work. Section III provides a description of the object detection approach. Section IV presents the results and analysis. Section V provides a summary of the work and scope for future work.

II. Related Work

The approaches for object detection on LiDAR point cloud can be classified as, Projection-based methods, Voxel-based methods, and Point-based methods [5].

The deep learning network in projection-based approaches take projection from different perspectives of point cloud as an input image. Point cloud can be projected either as Front View (FV) or as BEV. FVNet [7] and, VeloFCN [6] are the detectors built using FV projection. These methods can utilize image object detectors but face problems when occlusion occurs. BEV projection technique not only solves the occlusion problems but can also preserve depth information. Complex-Yolo [8], 3DYOLO [9], and [10] are some of the works utilizing this technique.

Detectors like VoxelNet [11] divide the point cloud into uniform cubes known as 3D Voxels. 3D CNN is applied to these voxels. 3D CNN makes it computationally more complex. Point-based techniques such as PointNet [12] use the network's raw point cloud as input straight away. The network receives the actual geometry and shape features, which is advantageous. Point-wise feature extraction makes these methods also time-consuming.

BEV-based techniques are faster than other approaches and also address the issue of occlusion. Complex-Yolo [8] and 3DYOLO [9] process 50.4 and 62 frames per second,

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respectively. In light of these benefits, this work uses a BEV-based technique for detection. Additionally, to further improve the detection accuracy, YoloV8 object detectors are used.

III. Methodology

The methodology entails employing the YOLOv8 model for 3D point cloud object detection, encompassing the extraction of BEV image data from Complex-YOLOv4 and subsequent model training and evaluation.

A. BEV Image Extraction

The focus of the research lies in the adaptation of the Complex-YOLO [8] architecture to detect objects within 3D point cloud data. Unlike traditional YOLO models that operate on 2D images, Complex-YOLO processes 3D point cloud data, making it suitable for tasks such as detecting objects in LiDAR data or other 3D sensing modalities. The Complex-YOLO architecture as shown in figure 1 is designed to process 3D point cloud data by transforming it into 2D bird's eye view (BEV) maps or images in the following manner. The point cloud is split into rectangular grids. Only pertinent regions of space, $X\epsilon[0, 50]$, $Y\epsilon[-25, 25]$, and $Z\epsilon[-2.73, 1.27]$, are taken into consideration to generate the BEV map in order to speed up the detection process. There are three RGB channels allocated to each grid or pixel. A number of 3D points are present in every pixel. Let a set of points be contained in a pixel P_i. The RGB channels are filled as follows:

$$r_i = min(1.0; log(N_i + 1)/64) \tag{1}$$

$$g_i = max(Intensity(P_i)) \tag{2}$$

$$b_i = max(P_i \cdot [0; 0; 1]^T) \tag{3}$$

Here, N_i denotes the number of points within pixel P_i. For pixel i, the point density is denoted by r_i, while the maximum of all Z values and intensity of all points in pixel i are represented by g_i and b_i, respectively. The above transformation allows for a simplified representation of the 3D data, making it easier to analyze and interpret. By converting the data into BEV format, Complex-YOLO facilitates object detection tasks, enabling the identification of objects from a top-down perspective.

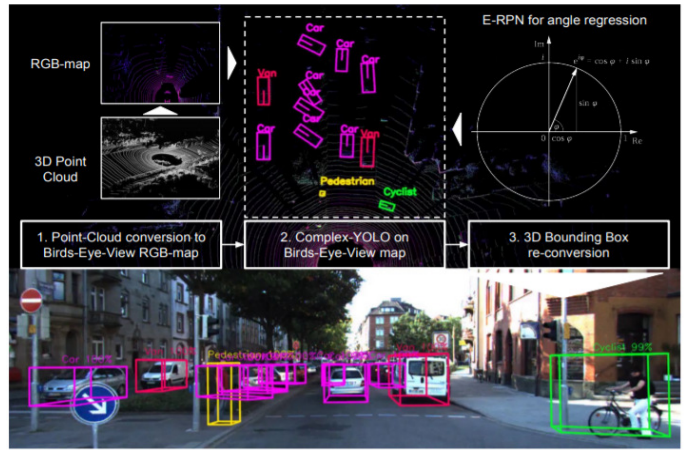

Fig. 1: Complex-yolo architecture [8]

To create a custom dataset for training the YOLOv8 model [14], 2D BEV data was extracted including images and their corresponding labels from the Complex-YOLOv4 model [16]. This extraction process allowed us to tailor the dataset to the specific requirements corresponding to YOLOv8. A total of 7140 images and their corresponding labels were extracted.

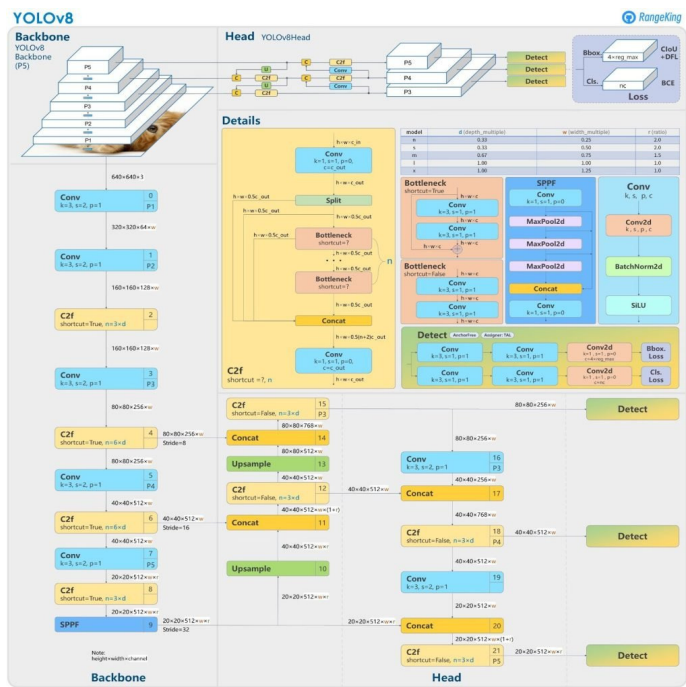

Fig. 2: YOLOv8 model architecture [15]

B. YOLOv8 Implementation

The YOLOv8 algorithm [14] represents a significant advancement in one-stage object detection, enabling faster detection compared to two-stage algorithms by predicting all possible edges simultaneously. Continuously updated and improved, the YOLO algorithm has evolved into the YOLOv8 version, which offers five different versions ranging from

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YOLOv8n to YOLOv8x, varying in depth and width of the network. YOLOv8 demonstrates improvements in detection speed and accuracy over previous versions such as YOLOv5 and YOLOv7. The YOLOv8 architecture as shown in Fig. 2 is based upon a modified version of the CSPDarknet53 backbone, comprising 53 convolutional layers interconnected via CSP connections to improve information flow. At the head of the network, numerous convolutional layers are followed by a sequence of fully connected layers. Together, these elements play a pivotal role in predicting bounding boxes, class probabilities, and objectness scores for identified objects within an image. A distinctive feature of YOLOv8 is that the head of the network incorporates a self-attention mechanism, adding further refinement to the detection process.The source code can refer to [14].

Following the extraction of 2D BEV data and their corresponding labels from the Complex-YOLOv4 architecture, the dataset underwent further processing to refine its structure and enhance its usability. One crucial aspect of this processing involved the modification of the label format to ensure compatibility with the training requirements of the subsequent YOLOv8 model. The original label format, comprising class ID, [x_{center}, y_{center}, $width$, $height$, and yaw] angle, was transformed into the YOLO oriented bounding box (OBB) format. This format adjustment involved converting the labels to include class ID and the coordinates of the four corners of the bounding box [xy_1, xy_2, xy_3, xy_4].

TABLE I: Detection results on Validation Split of KITTI Dataset

Class	Precision	Recall	AP
Car	0.933	0.891	0.953
Pedestrian	0.876	0.634	0.795
Cyclist	0.847	0.733	0.839

IV. Results and Analysis

The implemented object detector underwent testing using the extracted 2D Bird's Eye View (BEV) dataset from the KITTI Object dataset [13], which consists of a total of 7140 images further partitioned into 5712 training images and 1428 test images. An Intel Xenon processor and an NVIDIA GeForce RTX 3090 24 GB GPU powered the experiment's system. There were 100 training epochs used. The processing times were 1.1 ms for preprocessing, which was done before the data was fed into the model, 31.2 ms for inference, which is the amount of time the model takes to predict these outcomes for every image in the dataset, and 1.2 ms for postprocessing for each image. Table I displays the findings of the detection operation carried out on the evaluation split. The achieved mAP was 86.3%. The F1, Precision, Recall, and Confidence curves were obtained as shown in Fig. 4, Fig. 5, Fig. 6 and Fig. 7. A sample of model predictions is shown in Fig. 3.

The proposed detector was also tested on some of the point clouds from the TIAND dataset [17]. The qualitative result is shown in figure 8. The corresponding camera image is shown in figure 9. Indian traffic has several unstructured traffic environments. Technology Innovation Hub on Autonomous Navigation (TiHAN) IIT Hyderabad [18] is making an effort to create a multi-modal dataset specific to the Indian scenario. The dataset contains data from various sensor modalities: LiDAR, Camera, Radar, and GPS.

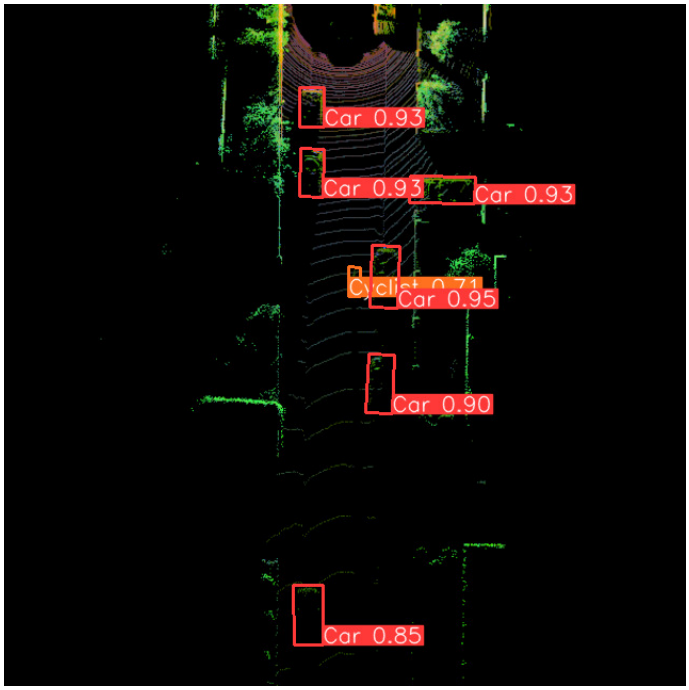

Fig. 3: Predictions on a BEV image retrieved from KITTI dataset [13]

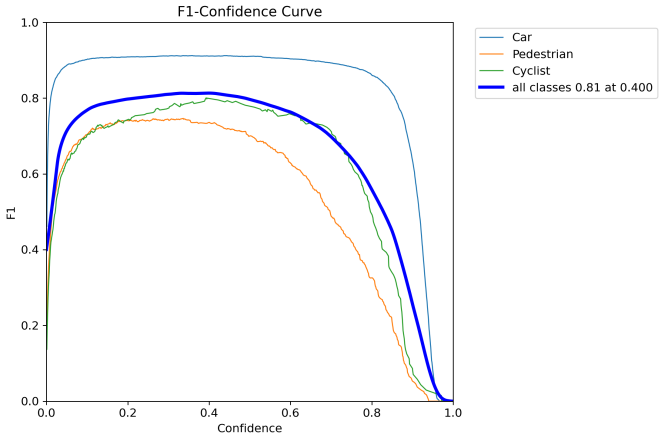

Fig. 4: F1-Confidence Curve

TABLE II: Comparison between YoloV4 based object detector and the proposed YoloV8 based detector

Detector	mAP (%)	FPS
YoloV4 based detection [10]	83.27	27.2
YoloV8 based detection (ours)	86.3	28.3

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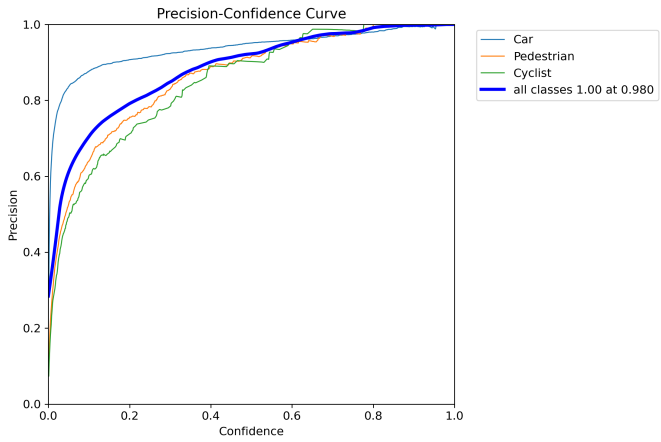

Fig. 5: Precision-Confidence Curve

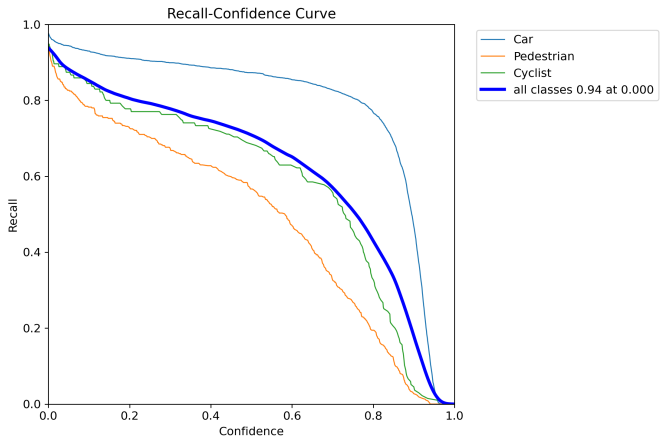

Fig. 6: Recall-Confidence Curve

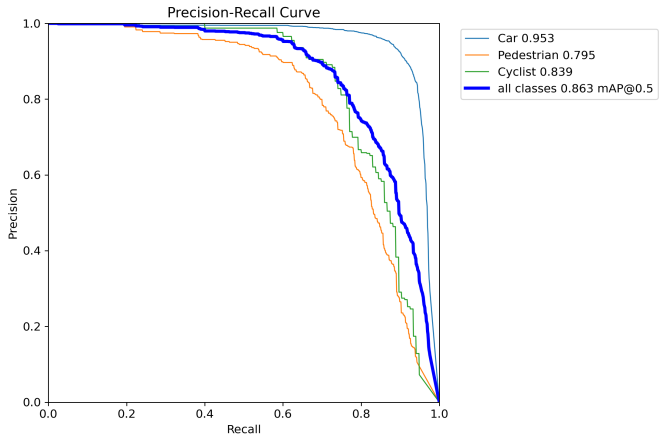

Fig. 7: Precision-Recall Curve

A comparison between the YoloV8-based detector proposed in this paper and the YoloV4-based detector [10] has been presented in table II. In comparison to [10] with an mAP of 83.27%, the proposed detector achieved an mAP of 86.3%, an

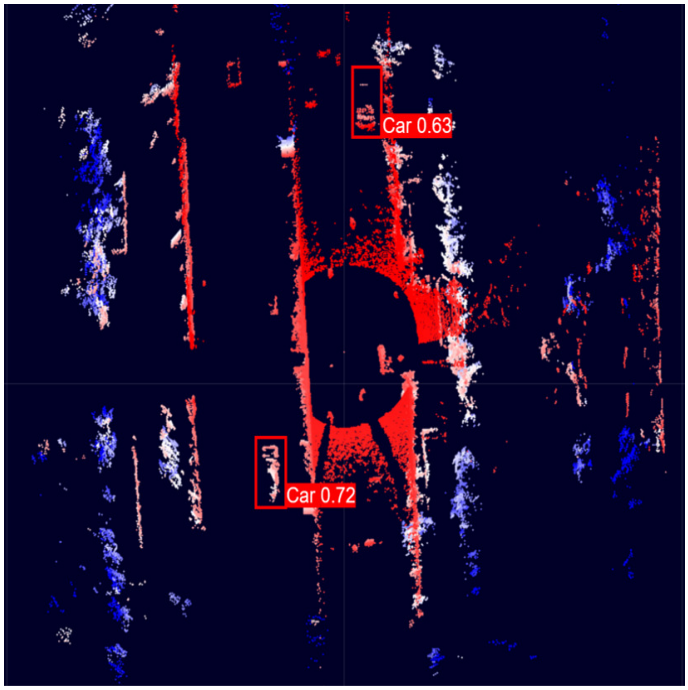

Fig. 8: Predictions on a BEV image retrieved from TIAND dataset [17]

Fig. 9: The camera image corresponding to the point cloud of figure 8 [17]

improvement of 3%. Also, the proposed detector processes the data at 28.3 Frames per Second (FPS), improving the inference time slightly.

V. Conclusion

This paper introduces a novel method for object detection in LiDAR data, employing the YOLOv8 network. By transforming the LiDAR point cloud into a pseudo-RGB image and

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feeding it into the YOLOv8 model, our approach achieves remarkable performance. Trained on the renowned KITTI dataset for classes including cars, pedestrians, and cyclists, our model achieves an mAP of over 86% on the validation split, with a rapid frame processing rate exceeding 28. This innovative model holds promise for enhancing environment perception with precise object detection, thereby offering valuable applications in ADAS. In the future, we aim to train the proposed model for various classes relevant to the Indian context and carefully analyze the obtained results.

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